INFORMATION TO KEEP WITH YOU

# My complete treatment handbook

## Pembrolizumab 200 mg · colorectal

Understand your treatment. Know what to expect. Keep your plan to hand.

**Displayed calendar**
**21 days**
See the active treatment phase

**Treatment stages**
**1**
Explained in the treatment reference

**Regimen source**
**NCCP 00455**
Version 16

Day 1: pembrolizumab 200 mg over about 30 minutes. Repeat every 3 weeks; allow extra time for checks.

## Treatment stages

01 **Pembrolizumab every 3 weeks**
One immunotherapy medicine

Your team confirms the active stage and dates. The full treatment guide explains each step.

## Inside your handbook


| | | PAGE |
|---|---|---|
| 01 | Understanding your treatment | 2 |
| 02 | Your schedule, tests and appointments | 5 |
| 03 | Symptoms and supportive care | 10 |
| 04 | Sources and further information | 16 |


**My name**

**Hospital / oncology team**

Use the urgent instructions and contact numbers in your treatment guide. Your team must replace example numbers before patient use. Your individual treatment plan takes precedence.

CHAPTER 01

# Understanding your treatment

IN THIS CHAPTER

Why this treatment is offered, what its medicines do, and how the agreed plan fits your care.

## When to get help

**Emergency: call 112 / 999**

Severe breathing difficulty, collapse, seizure or severe chest pain: call emergency services. Do not wait for an online answer.

**Contact your oncology service immediately**

Temperature 37.5°C or higher, 36°C or lower, shivering or suddenly feeling unwell — even with a normal temperature. Contact oncology immediately for new breathlessness, persistent diarrhoea, severe tummy pain, yellow skin or eyes, much less urine, spreading or blistering rash, severe headache, marked weakness, confusion, unusual thirst or passing much more urine. Severe breathing difficulty, chest pain, collapse or seizure needs emergency care. Effects can begin after treatment ends.

Follow the urgent contact instructions from your treating hospital. If you cannot reach the team, seek urgent medical assessment. Outside clinic hours, use your instructed out-of-hours pathway or the emergency department.

This page does not diagnose symptoms or decide whether it is safe to stay at home. You do not need to know which medicine caused a symptom before calling.

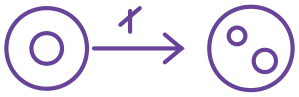
## Three medicines. Different jobs.

Pembrolizumab aims to control bowel cancer that has spread and has a particular deoxyribonucleic acid (the cell's genetic material) (DNA)-repair feature, called MSI-H or dMMR. Your team discusses likely benefit, uncertainty and alternatives for you.

IMMUNOTHERAPY

### Pembrolizumab

Keytruda



**Blocks an immune brake**

Pembrolizumab is an antibody that blocks PD-1, a control switch on immune T cells. Cancer cells can use this switch to reduce immune attack. Blocking it can help the immune system recognise and attack cancer. The same increased immune activity can inflame healthy organs, which is why new symptoms need attention even after treatment ends.

Immune cell → PD-1 blocked → Cancer attacked

**What to watch for**

Report new breathing, bowel, skin, hormone, heart or nerve symptoms promptly; these may be immune inflammation.

The drawings are simplified explanations of drug action, not anatomical pictures or a prediction of response.

## Your colorectal cancer and treatment aim

The tumour test and treatment setting matter.

### MSI-H and dMMR

These tests identify a problem with the cancer's DNA-repair system. They help identify colorectal cancers more likely to benefit from this immunotherapy. Ask whether genetics assessment is relevant to you.

### First-line metastatic treatment

Pembrolizumab aims to control bowel cancer that has spread and has a particular DNA-repair feature, called MSI-H or dMMR. Your team discusses likely benefit, uncertainty and alternatives for you. First-line means the first systemic treatment for the metastatic setting.

### Your choices

Ask about other treatments, clinical trials and symptom-focused care, how benefit will be assessed and what matters to your quality of life.

## How pembrolizumab works

It releases a brake on immune cells.

### Healthy tissues can be affected

The same immune activation can inflame the bowel, lungs, liver, kidneys, skin, hormone glands, heart, nerves or muscles. Some effects require hospital care, steroids or lasting hormone replacement.

### A pause is a safety decision

Doses may be delayed or treatment stopped for immune effects. The standard approach is to hold or stop treatment, not simply to give a smaller dose. Feeling better or having normal tests does not by itself mean that pembrolizumab can restart. Your team reviews what happened, recovery and the steroid or hormone plan. Some reactions mean that treatment must remain stopped, even if the symptoms have settled. Follow the prescribed plan and never restart treatment or change steroids yourself. Restart depends on the original reaction, recovery and any steroid or hormone plan. Some reactions mean treatment must remain stopped even after symptoms settle.

## Evidence in MSI-H/dMMR colorectal cancer

KEYNOTE-177 compared pembrolizumab 200 mg every three weeks with chemotherapy in previously untreated MSI-H/dMMR metastatic colorectal cancer.

The trial showed a longer time before cancer worsened with pembrolizumab. This is a group result, not a guarantee for one person; ask your team about benefits, risks and alternatives in your situation.

CHAPTER 02

# Your schedule, tests and appointments

IN THIS CHAPTER

Your treatment stages, checks and practical preparations, from before treatment through follow-up.

# Treatment → review → next steps

**Pembrolizumab 200 mg · colorectal**

21-day view




This shows treatment timing, not guaranteed symptom-free days. Your team confirms the active phase and dates.

ONE IMMUNOTHERAPY MEDICINE

**Pembrolizumab every 3 weeks**

C1 C2 C3 …

Continue while treatment helps and side effects remain acceptable. At 24 months, the team may consider stopping if there is a sustained response without progression; this is an individual review, not an automatic fixed final cycle.

Your team confirms each stage and discusses any changes with you.

BASELINE → TREATMENT → FOLLOW-UP

## Checks across your colorectal treatment

There is no fixed NCCP scan or tumour-marker interval in this regimen. The team individualises imaging, markers and extra tests.

The team arranges and reviews tests. This guide shows expected timing; it does not record completed appointments. NCCP 00455 v16 official regimen

## Tests in your plan

### Carcinoembryonic antigen (CEA) — colorectal plans

If part of your plan

WHEN

**For a colorectal cancer plan, if your team uses CEA; your team sets the interval.**

A marker trend interpreted alongside symptoms, examination and scans.

Record the baseline, latest result and next date. A marker change alone does not establish progression.

THE WHOLE TREATMENT JOURNEY

## Pembrolizumab 200 mg · colorectal: treatment and checks

**First-line metastatic MSI-H / dMMR colorectal cancer**

Pembrolizumab 200 mg is given over about 30 minutes on Day 1 every 3 weeks (21 days). The visit takes longer for assessment and preparation.

**Read from left to right.** Columns show stages, not equal lengths of time. This is a general plan: no dates or personal information are needed. Your team confirms which steps apply and arranges appointments.

**Treatment, tests and reviews across the course**

| Care / check | Before starting | Infusion visits | Between visits | Course review | After treatment |
|---|---|---|---|---|---|
| **Blood count, kidneys, liver and glucose** NCCP 00455 | Baseline blood tests | Before every treatment cycle | Extra tests for new symptoms or abnormalities | Clinical review before restart after a hold | Follow abnormal results to resolution |
| **Thyroid and other hormones** NCCP 00455 | Baseline thyroid function | Check thyroid timetable | Thyroid every 3–6 weeks; cortisol if indicated | Confirm any hormone replacement plan | Hormone treatment may continue long term |
| **Symptoms and immune inflammation** NCCP 00455 | Discuss autoimmune disease, infection, medicines and previous immunotherapy | Review new symptoms before infusion | Report symptoms immediately; do not wait for the next cycle | Pauses or stopping are clinician decisions | Immune effects can occur after the last dose |
| **Cancer response and duration** NCCP 00455 | Confirm validated MSI-H/dMMR result and first-line setting | Agree the response assessment plan | Scans and markers as directed by the consultant | Individual review, including possible stop after sustained response at 24 months | Agree follow-up and the contact pathway |

## What each check is for

### 1 Pembrolizumab every 3 weeks

One immunotherapy medicine

Pembrolizumab aims to control bowel cancer that has spread and has a particular DNA-repair feature, called MSI-H or dMMR. Your team discusses likely benefit, uncertainty and alternatives for you.

**Ask / act:** Confirm the current phase, prescription, tests and support medicines.

**At the next review, ask**

- Which treatment phase am I in, and what happens next?
- Which test is expected next, who arranges it, and who explains the result?
- Who should I contact if an expected appointment has not arrived?

This map does not show whether a test has been ordered, booked or completed. Do not wait for a scheduled test if you develop new symptoms.

## Practical treatment planning

Agree the arrangements with your oncology team.

STEP 1

### Appointments

STEP 2

### Course length

Continue while treatment helps and side effects remain acceptable. At 24 months, the team may consider stopping if there is a sustained response without progression; this is an individual review, not an automatic fixed final cycle.



STEP 3

### Blood and thyroid appointments

Blood count, kidney/liver function, glucose and thyroid function at baseline. Blood count, kidney/liver and glucose before each 3-weekly cycle; thyroid every 3–6 weeks regardless of infusion interval; cortisol if indicated. Scans and markers follow the consultant plan.



STEP 4

### Home medicines and records

There is no take-home chemotherapy pump. Keep your alert card, symptom record and medicine list. Discuss travel, work and transport needs.

STEP 5

### After the final dose

Keep reporting new symptoms. Ask who follows persistent side effects and any hormone medicines.

## Your treatment course

Visit frequency and total course length are separate decisions.

### Evidence for the schedule

KEYNOTE-177 tested 200 mg every three weeks. The 400 mg six-week schedule is supported by exposure bridging and product authorisation; it was not a separate randomised colorectal comparison.

## Tests and follow-up

Blood tests can identify a problem before symptoms appear.

### Scans and tumour markers

The consultant sets the disease-assessment plan. This regimen does not prescribe a universal computed tomography (CT) or carcinoembryonic antigen (CEA) interval. A scan schedule from a research study is not automatically your schedule.

### During a hold and after stopping

Keep follow-up appointments and the agreed steroid or hormone plan. Continue reporting symptoms after the last dose; another clinician needs to know you have received immunotherapy.

CHAPTER 03

# Symptoms and supportive care

IN THIS CHAPTER

Recognising changes, knowing when to call, and using your prescribed supportive-care plan.

## Your 21-day cycle.

Timing is a general guide; symptoms may start or stop at other times.

| Treatment or check | Timing in this view |
| --- | --- |
| **Infusion** |  <br>**Day 1**<br>Day 1 · 200 mg infusion |
| **Immune effects** |  <br>**Days 1–21**<br>Any day, and after the final dose |
| **Symptoms and thyroid checks** |  <br>**Days 1–21**<br>Thyroid tests every 3–6 weeks; report symptoms immediately |

The cycle shows appointment timing. It does not show a predictable immune-toxicity peak.

The next cycle starts 21 days after this one, if your team confirms it.

> Immune effects can happen on any day and after treatment ends. Carry your immunotherapy alert card and tell any healthcare team that you receive pembrolizumab.

### Between-treatment notes

### Day 1: Infusion and assessment

The team reviews blood results, symptoms and medicines. Tell the nurse immediately about itching, flushing, tight chest, dizziness or suddenly feeling unwell. Confirm the next tests, treatment visit and contact plan.

### Day 2–21: Between infusions

Immune effects may start at any time. A long interval between treatment visits is not a break from symptom reporting. Keep thyroid tests every 3–6 weeks, including separate appointments if required.

## Support around the medicines

1

### Routine medicines

Routine premedication is not usually required. Pembrolizumab has minimal emetogenic potential; the team individualises anti-sickness treatment if needed.

### 2 Steroids and hormone replacement

If immune effects develop, the team may prescribe steroids or hormones. Follow the exact plan; do not stop steroids abruptly or restart pembrolizumab yourself. If treatment is paused because of an immune side effect, your team checks recovery and any steroid or hormone plan before deciding whether it can restart. Do not change steroids or restart treatment yourself.

### 3 Medicines and pregnancy

Bring prescription, non-prescription and herbal medicines. Discuss pregnancy prevention; effective contraception is required during pembrolizumab and for at least four months after the last dose. Do not breastfeed during treatment.

## Eating, energy and daily life

Report changes early.

### Bowel changes

Report new or persistent diarrhoea, blood in stools or abdominal pain promptly. Do not treat ongoing diarrhoea as a minor stomach upset or rely on self-treatment without advice.

### Energy and hormones

New marked fatigue, headache, dizziness, feeling unusually cold, unusual thirst or passing more urine may need hormone and glucose tests.

### Medicines, pregnancy and feeding

## When to get urgent help

Do not wait for the next infusion.

### During the infusion

Tell the nurse immediately about itching, flushing, breathlessness, chest tightness, dizziness or feeling suddenly unwell.

### Carry the alert card

Show your treatment card to any healthcare team, even after immunotherapy ends. Keep your actual daytime and out-of-hours oncology numbers accessible.

**When heart, muscle and nerve symptoms overlap**

Heart, muscle and nerve problems can occur together. New muscle pain or weakness with changed vision, unusual tiredness, breathlessness, chest pain or an irregular heartbeat needs immediate medical advice. Do not wait for several symptoms to appear together. Severe breathing difficulty, collapse or rapidly worsening weakness needs emergency care.

## Questions worth asking

**undefined**

undefined

**undefined**

undefined

**undefined**

undefined

**undefined**

undefined

**undefined**

undefined

**undefined**

undefined

**Try explaining the plan back:** Which medicines are you having now? What changes after induction? What would make you call today? Ask your team to fill any gaps.

---

**Colorectal treatment context** This guide covers first-line metastatic MSI-H/dMMR colorectal cancer. The shared NCCP regimen has other cancer indications, but their tests, treatment aims and durations are not this guide.

**Supportive prescription draft for review.** This is a regimen template. It is not an issued patient prescription; individual review and completion are required.

## Pembrolizumab 200 mg every 3 weeks

**Active medicines:** pembrolizumab

**Emetogenic risk:** Minimal

No routine prophylactic antiemetic order is populated for this phase. Agree an individual symptom-treatment plan where needed.

### Additional phase-specific support

- No routine antiemetic prophylaxis or routine infusion premedication is usually required. Use an individual/local symptom-relief prescription if needed.

- Carry the immunotherapy alert card and report possible immune symptoms promptly during and after treatment.
- Effective contraception during treatment and at least 4 months after the last dose.
- For this colorectal context, eligibility is first-line MSI-H/dMMR metastatic disease; PD-L1 is not the colorectal selection test. Do not import a lung adjuvant course cap.
- full blood count (FBC), renal/liver profile and glucose before each cycle. Thyroid tests every 3–6 weeks: keep thyroid checks when due regardless of the infusion interval. Cortisol as clinically indicated.
- Review any immune toxicity and steroid dose before restarting; no routine dose reduction. Delayed immune effects may occur after treatment stops.
- If treatment is paused because of an immune side effect, your team checks recovery and any steroid or hormone plan before deciding whether it can restart. Do not change steroids or restart treatment yourself.

**Before signing:** verify allergies, current medicines, interactions, renal / hepatic function, QT risk, sedation, prior response, total steroid exposure and the rescue-medicine duration. Do not duplicate medicines from another phase.

Individual changes / rescue duration: _____
Prescriber / date: _____ Pharmacy review: _____

# What to watch for

You do not need to know which medicine caused a symptom before calling.

### 1 Hormones / nervous system

**Pembrolizumab**

Severe headache, confusion, weakness, dizziness or marked thirst need prompt advice.

### 3 Heart

**Pembrolizumab**

Chest pain, racing heartbeat or fainting needs urgent assessment.

### 5 Kidneys

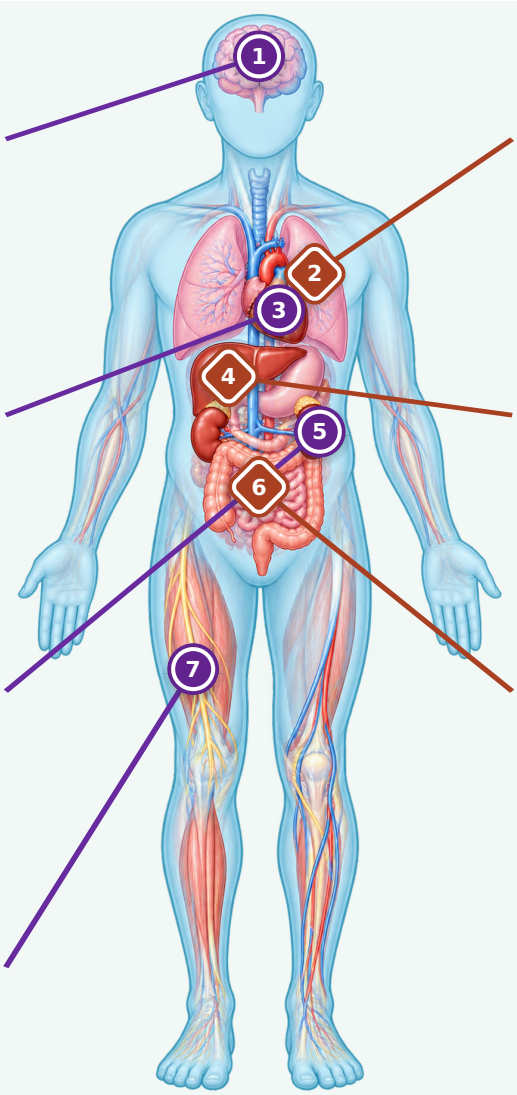
**Pembrolizumab**

Report much less urine, new swelling or a change in kidney tests.

### 7 Skin / muscles

**Pembrolizumab**

Spreading rash, blisters, severe weakness or muscle pain needs prompt review.




### 2 Lungs

**Pembrolizumab**

New cough or breathlessness can signal inflammation or another urgent cause.

### 4 Liver

**Pembrolizumab**

Yellow eyes/skin or dark urine may signal inflammation; tests also check the liver.

### 6 Bowel

**Pembrolizumab**

New or persistent diarrhoea, blood in stools or severe tummy pain needs urgent advice.

Selected effects, not every possible effect or a prediction. Blood-count, skin and line markers are schematic. Number and text carry meaning; colour is decorative.

Contact oncology immediately for new breathlessness, persistent diarrhoea, severe tummy pain, yellow skin or eyes, much less urine, spreading or blistering rash, severe headache, marked weakness, confusion, unusual thirst or passing much more urine. Severe breathing difficulty, chest pain, collapse or seizure needs emergency care. Effects can begin after treatment ends.

**When to get help**

**Call 112 / 999** for severe breathing difficulty, severe chest pain, collapse or a seizure.

**Call oncology immediately:** temperature 37.5°C or above, 36°C or below, shivering or suddenly feeling unwell, even with a normal temperature. Follow your hospital's temperature instructions. If you cannot reach the team, seek urgent medical assessment. Do not wait for the next appointment.

**Oncology daytime:** 087 12345678 · **Out of hours:** 087 87654321

**EXAMPLE ONLY — team must replace both numbers before patient use.**

CHAPTER 04

# Sources and further information

## IN THIS CHAPTER

The references behind this information and addresses you can use to find further help.

Use this information alongside the plan agreed with your treatment team. Your team confirms your medicines, doses and appointments.


1. **NCCP 00455 v16 · pNCCP00455 pp5–7**
   https://healthservice.hse.ie/documents/6903/Pembrolizumab_200mg_Monotherapy_455.pdf
2. **EMA Keytruda intravenous (IV) product information and patient leaflet**
   https://www.ema.europa.eu/en/documents/product-information/keytruda-epar-product-information_en.pdf
3. **KEYNOTE-177 randomised first-line colorectal trial**
   https://pubmed.ncbi.nlm.nih.gov/33264544/
4. **KEYNOTE-177 final analysis**
   https://pubmed.ncbi.nlm.nih.gov/35427471/
5. **Pembrolizumab 400 mg six-week exposure bridging**
   https://pubmed.ncbi.nlm.nih.gov/32305010/
6. **HSE treatment safety and urgent advice**
   https://www2.hse.ie/conditions/chemotherapy/side-effects/
7. **NCI: colon cancer treatment — carcinoembryonic antigen (CEA) and follow-up**
   https://www.cancer.gov/types/colorectal/hp/colon-treatment-pdq